Class D Audio Amplifier

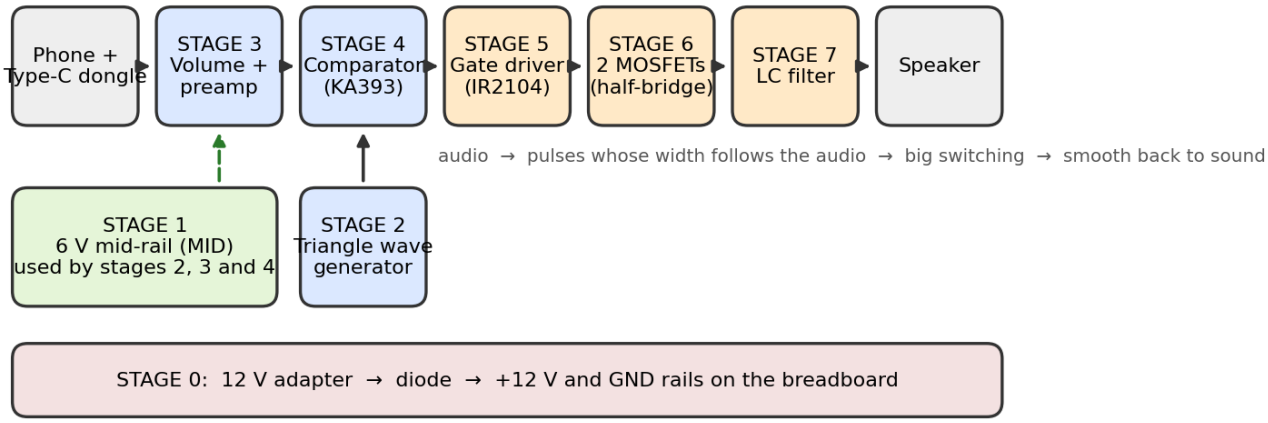
Beginner build guide for a breadboard — no experience needed

What you are building

This circuit takes music from your phone and plays it through a speaker. A normal amplifier makes the signal bigger smoothly, and wastes a lot of power as heat. This one works differently: it turns the music into very fast ON/OFF pulses (about 55,000 per second), lets two switches (MOSFETs) handle the big current, and then a small filter smooths the pulses back into music. Switches barely get warm, which is why this type of amplifier is efficient.

You will build it in **8 small stages (0 to 7)**. After every stage there is a quick measurement to prove it works. **Do not skip the tests**: they tell you exactly which stage has a mistake, so you never have to hunt through the whole board.

How the amplifier works, left to right (build in the order of the stage numbers)



Words used in this guide

Word	Meaning
+12 V	The positive supply from your adapter. On the breadboard it is the long **red** line at the top.
GND (ground)	The 0 V line that everything returns to. The long **blue** lines. All grounds must be joined together.
MID	A 6 V line that the circuit makes for itself, halfway between 0 V and 12 V. The op-amp chips need it as their "zero", because they only get one 12 V supply.
Pin	One metal leg of a chip. Pins are counted from pin 1 (see the pinout page).
Column	On a breadboard, 5 holes in a short vertical line are joined together inside. Two things plugged into the same column are connected.
Jumper	A short piece of wire used to connect two columns.
Decoupling cap	A small 0.1 µF capacitor placed right across a chip's power pins. It soaks up electrical spikes. Always add it, never leave it out.
PWM	Pulses of the same height whose width follows the music: louder music means wider pulses.

Before you start

SAFETY	1) Never add, move or remove a wire while the adapter is plugged in. 2) Electrolytic capacitors and diodes have a direction: backwards ones can pop or burn. 3) Keep the speaker disconnected until Stage 7. 4) If anything gets hot, smells, or the 10 Ω resistor warms up, unplug immediately and re-check the wiring. 5) Test after each stage, before building the next.

Tools

- A digital **multimeter** (set to DC volts, 20 V range) — this is your main test tool.
- An **oscilloscope** from your lab, if you can use one: it shows the waves directly. Without it you can still follow the voltage tests, but you will not see the waves.
- Jumper wires (or solid 22 AWG wire and wire strippers), tweezers or small pliers, a pencil and paper to tick off each step.

Reading the part markings

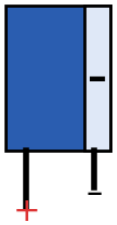
Part	How to recognise it
10 kΩ resistor	brown – black – orange – (gold)
22 kΩ resistor	red – red – orange – (gold)
1 kΩ resistor	brown – black – red – (gold)
10 Ω resistor	brown – black – black – (gold)
100 kΩ trimmer	small blue/white box with a screw, printed **104**
Small ceramic capacitors	printed **102** = 1 nF, **104** = 0.1 µF, **105** = 1 µF
Big capacitors	Have a stripe with a "–" sign. The stripe leg is the minus; the longer leg is the plus.

If you are not sure about a resistor, measure it: put the multimeter on Ω (resistance) and touch the two legs.

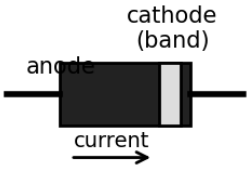
Parts that have a direction



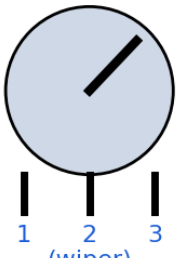
MOSFET legs:
Gate – Drain – Source



Electrolytic cap:
stripe = minus,
longer leg = plus



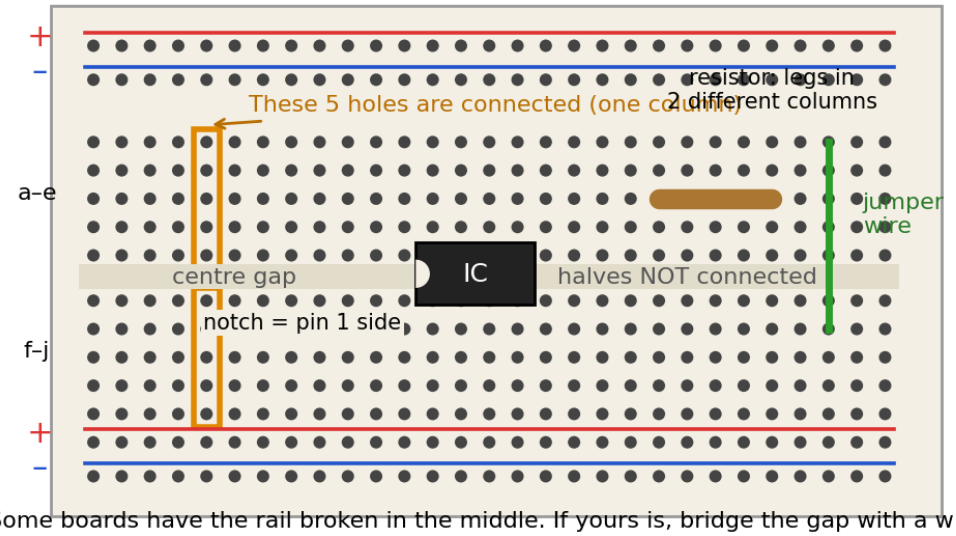
Diode: band side is
the cathode (K)



Pot / trimmer: 3 legs,
middle = wiper

How a breadboard connects

Rails run along the whole length: top red = +12 V, top blue = GND

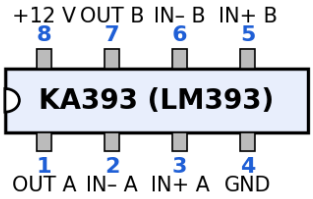


Some boards have the rail broken in the middle. If yours is, bridge the gap with a wire.

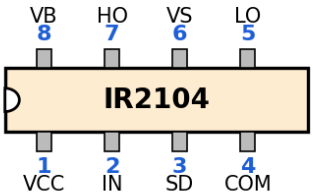
Pin numbers of the three chips

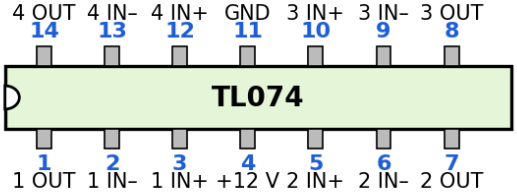
Every chip has a small notch (or dot) at one end. Place the chip so the notch is on your **left**. Pin 1 is then at the bottom left. Count along the bottom row going right, then continue along the top row going left. **Straddle every chip across the centre gap** so the left and right pins are not shorted together.

Chips seen from above, notch on the LEFT (as on your breadboard)



Pin 1 = bottom-left.
Count along the bottom row,
then go up and come back
along the top row.

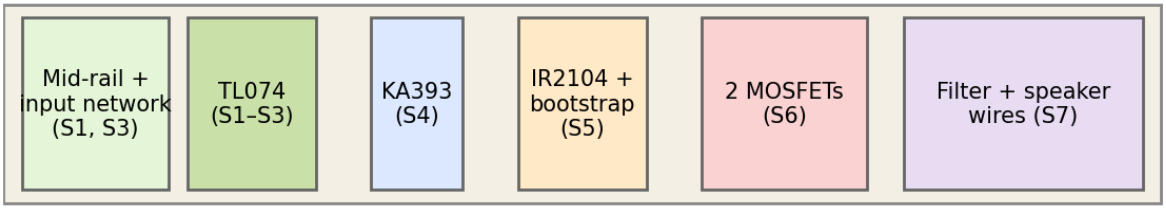




TL074 sections: A = pins 1,2,3 B = pins 5,6,7 C = pins 8,9,10 D = pins 12,13,14

Where to put things on the board

Suggested left-to-right placement on ONE 830-point breadboard (about 63 columns)



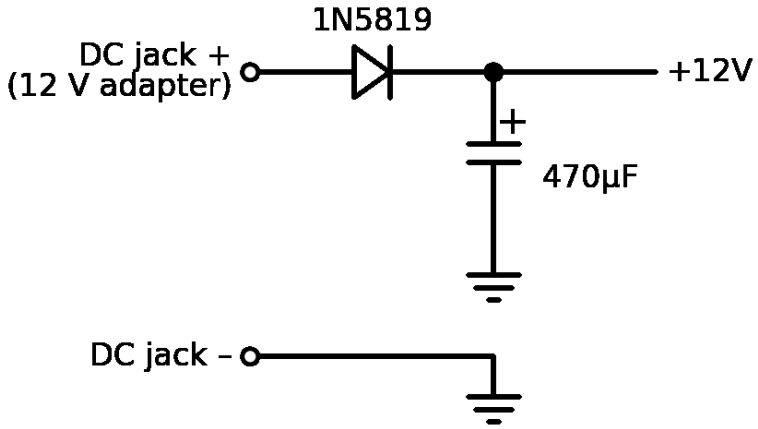
Leave 3 or more empty columns between chips so you have room for wires

Tip: use **red** wires for +12 V, **black** wires for GND, **orange** for MID, and yellow or green for signals. Wire colours make mistakes very easy to spot.

STAGE 0 — Power on the breadboard

What it does: Brings 12 V from your adapter onto the long rails. The diode protects the circuit if you plug the adapter in backwards, and the big 470 µF capacitor keeps the voltage steady.

You need: 12 V adapter with a barrel-jack breakout (screw terminals), 1N5819 diode, 470 µF capacitor, one 10 Ω resistor (temporary, a safety "fuse" for the first tests), jumper wires.



Connections — do them in this order

1. Plug the adapter into the barrel-jack breakout. With the multimeter on DC volts, touch the two screw terminals: the red probe on **+** gives a positive number (about 12–15 V). Unplug the adapter again and mark which terminal is +.
2. On the breadboard, join the top blue rail and the bottom blue rail with a wire. Both are GND. Do **not** connect the top red rail to the bottom red rail: the bottom red rail will be MID.
3. Adapter **– terminal** → top blue rail.
4. Adapter **+ terminal** → one leg of the 10 Ω resistor. Other leg of the 10 Ω → the **plain (unbanded) end** of the 1N5819 diode.
5. The **banded end** of the diode → top red rail (+12 V).
6. Put the 470 µF capacitor between the rails: the leg next to the **stripe (–)** goes into the blue rail, the other leg goes into the red rail.

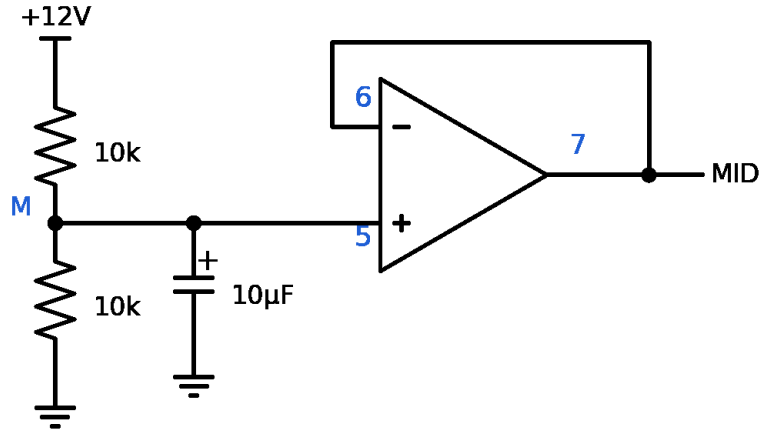
TEST	Plug in the adapter. Put the multimeter on the top red rail (red probe) and top blue rail (black probe). You should read about **11 to 14 V**. Unplug the adapter before you continue, and do this every time you add parts.

If the test fails: a reading of 0 V or a negative number means the diode, adapter polarity or the capacitor is the wrong way round. Re-check steps 3 to 6.

STAGE 1 — The 6 V mid-rail

What it does: The chips only have a 12 V supply, so they need a 6 V "zero line" for the audio to swing above and below. Two equal resistors split 12 V into 6 V, and one section of the TL074 chip copies that voltage so it can feed several places without sagging.

You need: TL074 chip, two 10 kΩ resistors, one 10 µF electrolytic capacitor, one 0.1 µF capacitor.



Connections — do them in this order

1. Push the **TL074** across the centre gap, notch on the left (use columns 12 to 18). Press down evenly.
2. Power it: **pin 4 → red rail (+12 V)** and **pin 11 → blue rail (GND)**. These two pins power all four sections of the chip.
3. Put a **0.1 µF** capacitor across pin 4 and pin 11 (decoupling).
4. Choose a free column and call it **M**. Put a 10 kΩ resistor from the red rail to M, and a second 10 kΩ from M to the blue rail.
5. Put the **10 µF** capacitor with its **+ leg in column M** and its stripe (–) leg in the blue rail.
6. Jumper wire: column M → **TL074 pin 5**.
7. Jumper wire: **pin 6 → pin 7**.
8. Jumper wire: **pin 7 → bottom red rail**. That rail is now **MID** (6 V). Put a piece of tape or a label on it.

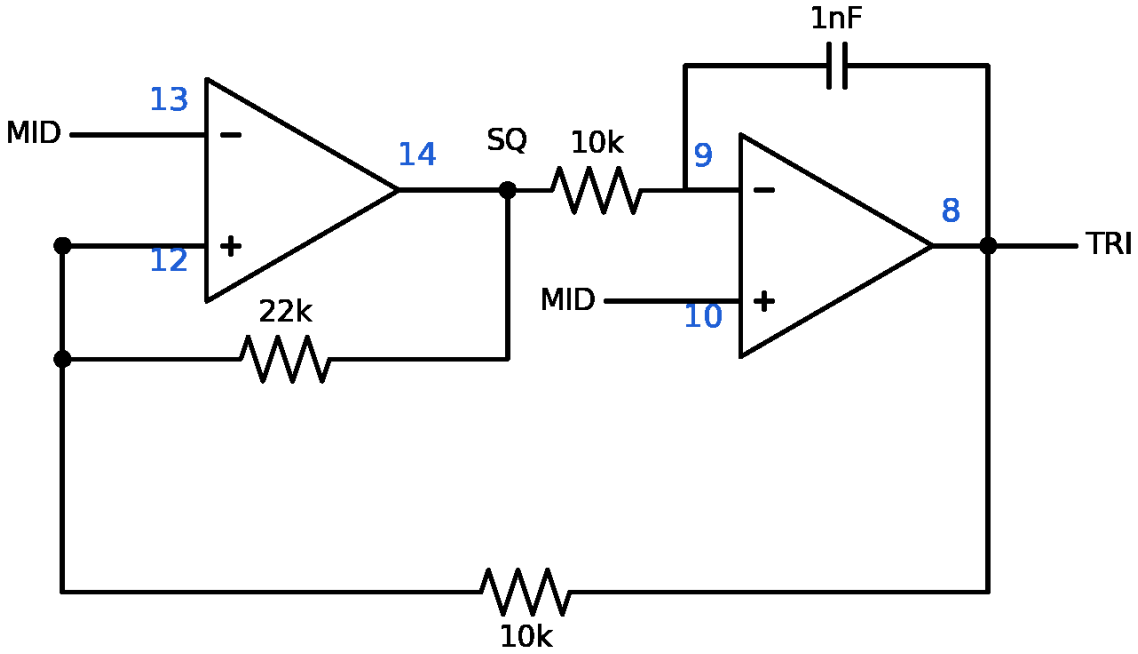
TEST	Plug in. Black probe on GND (blue rail), red probe on the **bottom red rail**: you should read **about 6 V** (5.5 to 6.5 V). Also read pin 7: it must show the same.

If the test fails: if you read 0 V or 12 V, check TL074 pins 4 and 11 first (power), then pins 5, 6 and 7.

STAGE 2 — The triangle wave generator

What it does: Makes a triangle wave (a signal that rises and falls in straight ramps) about 55,000 times a second. It is the "ruler" that the music is later measured against. It uses two more sections of the same TL074: one flips between high and low (Schmitt trigger) and the other turns that into ramps (integrator).

You need: The same TL074 chip, two 10 kΩ resistors, one 22 kΩ resistor, one 1 nF capacitor (marked 102).



Connections — do them in this order

1. **Pin 10 → MID** (bottom red rail).
2. **Pin 13 → MID**.
3. 10 kΩ resistor between **pin 14 and pin 9**.
4. 1 nF capacitor between **pin 8 and pin 9**.
5. 10 kΩ resistor between **pin 8 and pin 12**.
6. 22 kΩ resistor between **pin 14 and pin 12**.

Tip: each pin has its own column. Put one leg of a part in the same column as the pin (on its side of the gap), and the other leg in the column it must reach.

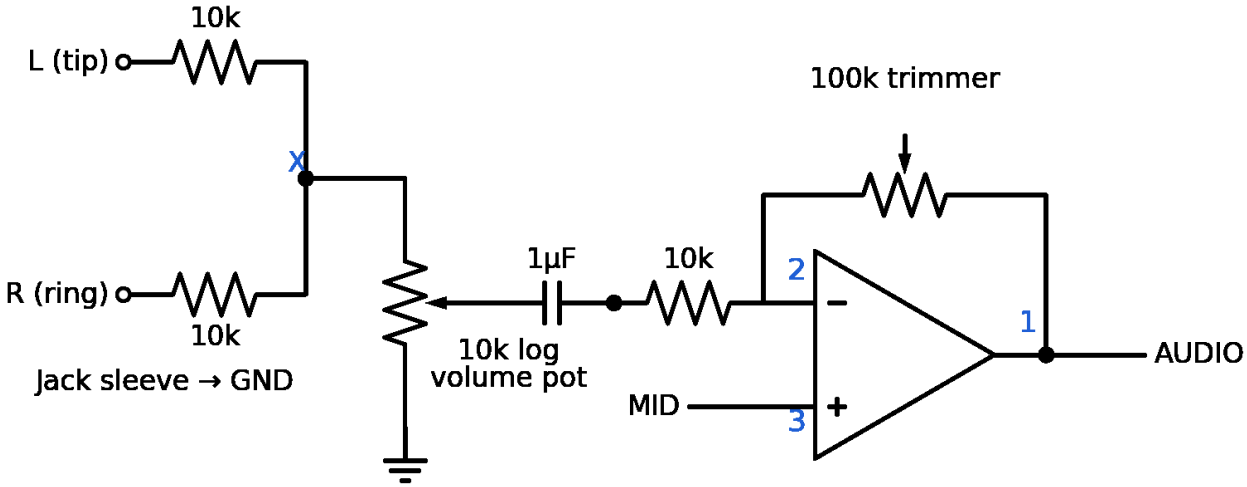
TEST	**With an oscilloscope:** probe on pin 8 (ground clip on the blue rail): a triangle wave about 4 V peak-to-peak, centred on 6 V, repeating roughly every 18 µs (around 55 kHz; anything between 40 and 70 kHz is fine). Pin 14 shows a square wave. **With only a multimeter:** pin 8 and pin 14 should both read about 6 V.

If the test fails: if pin 8 sits stuck near 1 V or 10 V, one of pins 9, 12 or 13 is connected wrongly. Compare with the diagram pin by pin.

STAGE 3 — Phone input, volume and preamplifier

What it does: Mixes the left and right audio from the phone, lets you set the volume with the pot, and then boosts the signal with the last section of the TL074. The 1 µF capacitor lets the music through but blocks the 6 V DC level. The trimmer sets the boost between 0 and 10 times.

You need: 3.5 mm audio jack breakout (or an old aux cable cut open), Type-C to 3.5 mm adapter, 10 kΩ log pot, 100 kΩ trimmer, three 10 kΩ resistors, one 1 µF capacitor.



Connections — do them in this order

1. **TL074 pin 3 → MID**.
2. Trimmer: join its **centre leg to its left leg** with a short wire. Then **left leg → TL074 pin 1** and **right leg → TL074 pin 2**. Turn the screw to the middle.
3. 10 kΩ resistor from **pin 2** to a free column called **P**.
4. 1 µF capacitor from column **P** to the **middle leg of the volume pot**. (If it is an electrolytic type, put + on the P side.)
5. Volume pot: **right leg → blue rail (GND)**, **left leg → column X** (a new free column).
6. Two 10 kΩ resistors: one from column X to the jack's **tip** (left channel), one from column X to the jack's **ring** (right channel). The jack's **sleeve** (the long base part) goes to the blue rail (GND).
7. If you use a cut aux cable: identify its three wires (tip, ring, sleeve) by plugging it in and testing continuity with the multimeter on the beep range, touching each wire to the plug's tip, middle ring and base.

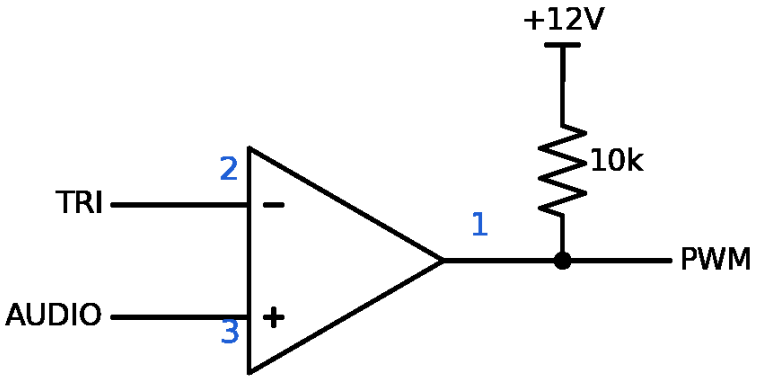
TEST	Plug the dongle into your phone, the 3.5 mm plug into your jack, and play music at low volume. Power on and put the multimeter on **pin 1**: it should read **about 6 V** and stay steady. With an oscilloscope on pin 1 you will see the music as a small wiggly line centred on 6 V.

If the test fails: if pin 1 is near 0 V or 12 V, check pin 2, pin 3 and the trimmer wiring. If the volume is backwards (turning up makes it quieter), swap the two outer pot legs.

STAGE 4 — The comparator (makes the pulses)

What it does: The comparator is a decision maker. It looks at the music and the triangle wave and says YES (high) when the music is above the triangle, and NO (low) when it is below. The result is a train of pulses whose width follows the music. That is PWM.

You need: KA393 chip (it is the same as an LM393), one 10 kΩ resistor, one 0.1 µF capacitor.



Connections — do them in this order

1. Push the **KA393** across the centre gap, notch on the left, at least 3 columns away from the TL074.
2. **Pin 8 → red rail (+12 V)**, **pin 4 → blue rail (GND)**, and a **0.1 µF** capacitor across pins 8 and 4.
3. Jumper wire: **KA393 pin 2 → TL074 pin 8** (the triangle wave).
4. Jumper wire: **KA393 pin 3 → TL074 pin 1** (the audio).
5. **10 kΩ resistor from KA393 pin 1 to the red rail (+12 V).** This chip can only pull its output down, so this resistor pulls it up.
6. The second half of the chip is not used: connect **pin 5 → GND** and **pin 6 → MID**. Leave pin 7 empty.

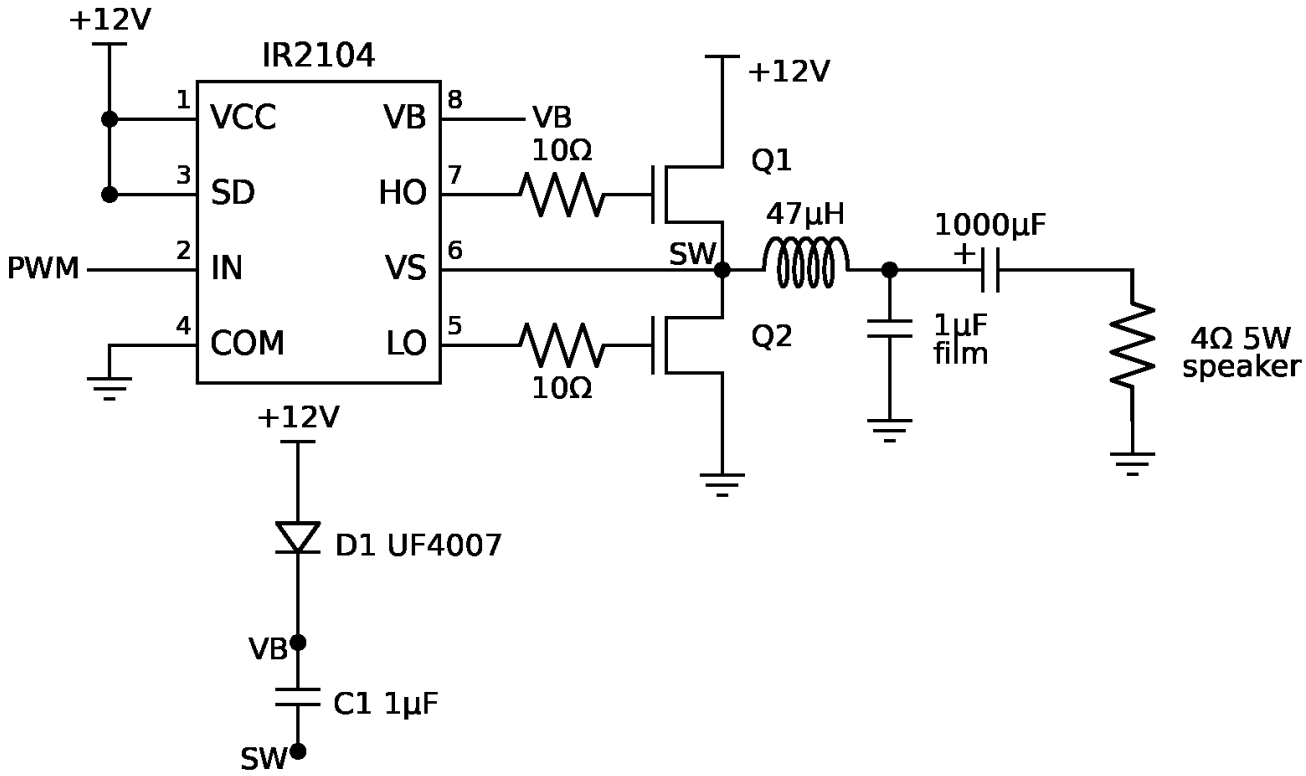
TEST	**Oscilloscope** on KA393 pin 1: square pulses between about 0 V and 12 V at the triangle's frequency. Play music: the pulse widths wobble. **Multimeter only**: pin 1 reads about 6 V when the music is silent, and the reading moves up and down when music plays.

If the test fails: if pin 1 is stuck at 12 V or 0 V, check that pins 2 and 3 are not swapped and that the 10 kΩ pull-up is in place.

STAGE 5, 6 and 7 — The output half: driver, MOSFETs and filter

Stages 5, 6 and 7 are all on this one diagram. **Build them one at a time** and test after each, as described on the next pages.

Gate driver, half-bridge output stage and LC output filter



Remember: the 10 Ω "safety" resistor between the adapter and the diode (Stage 0) stays in until the end of Stage 6. The speaker stays disconnected until Stage 7.

STAGE 5 — The gate driver (IR2104)

What it does: MOSFETs need a strong push on their gate to switch quickly. The IR2104 chip takes the pulses and creates two gate signals: HO for the top MOSFET and LO for the bottom one, with a tiny pause between them so both are never on together. The diode and the 1 µF capacitor on pin 8 let it create a voltage higher than 12 V for the top MOSFET (this trick is called a bootstrap).

You need: IR2104 chip, UF4007 diode, one 1 µF capacitor (ceramic or film, 25 V or more), two 10 Ω resistors, one 0.1 µF capacitor.

Connections — do them in this order

1. Push the **IR2104** across the centre gap, notch on the left, away from the other chips.
2. **Pin 1 → red rail (+12 V)**. **Pin 4 → blue rail (GND)**. A **0.1 µF** capacitor across pins 1 and 4.
3. Jumper: **pin 3 → pin 1**. This switches the chip on (a low voltage on pin 3 would shut it down).
4. Jumper: **pin 2 → KA393 pin 1** (the pulses).
5. **UF4007** diode: the **plain end → red rail (+12 V)** and the **banded end → IR2104 pin 8**.
6. **1 µF capacitor between pin 8 and pin 6.** (If it is an electrolytic, + goes to pin 8.)
7. 10 Ω resistor from **pin 7** to a free column, called **Q1G**.
8. 10 Ω resistor from **pin 5** to another free column, called **Q2G**.

TEST	No MOSFETs yet. Power on: the multimeter on **pin 5** reads about 6 V when silent, and **pin 2** also about 6 V. With an oscilloscope on pin 5, you see 0 to 12 V pulses (an upside-down copy of the pulses on pin 2).

If the test fails: if pin 5 is stuck at 0 V, check pins 1, 3 and 4. A stuck pin 5 with pin 3 left floating is the most common mistake.

STAGE 6 — The two MOSFETs (the switches)

What it does: Q1 connects the point called SW to +12 V, and Q2 connects SW to ground. They take turns very quickly. The longer Q1 stays on, the higher the average voltage at SW, so SW follows the music.

You need: Two IRLZ44N MOSFETs, one 0.1 µF capacitor.

Connections — do them in this order

1. Hold a MOSFET with its **printed side facing you** and the legs pointing down. The legs from left to right are **Gate, Drain, Source**. Plug each MOSFET into three separate columns.
2. **Q1** (top switch): Gate → column **Q1G**. Drain → **red rail (+12 V)**. Source → a free column called **SW**.
3. **Q2** (bottom switch): Gate → column **Q2G**. Drain → column **SW**. Source → **blue rail (GND)**.
4. Jumper: **IR2104 pin 6 → column SW**.
5. Put a 0.1 µF capacitor between the red and blue rails right next to the MOSFETs.

TEST	Speaker still disconnected. Power on and measure the voltage on column **SW**: about **6 V** when silent. The MOSFETs should stay cool or only slightly warm. The 10 Ω resistor must **not** get warm.

If the test fails: if SW reads 0 V or 12 V constantly, or the 10 Ω resistor warms up, unplug at once. Check that Drain and Source of each MOSFET are on the right columns, that pin 3 of the IR2104 is connected to pin 1, and that the diode is the right way round.

STAGE 7 — Output filter and speaker

What it does: The inductor and the 1 µF capacitor form a filter: they let the slow music through and smooth away the fast 55 kHz switching. The big 1000 µF capacitor blocks the 6 V DC level so only the music reaches the speaker.

You need: 47 µH power inductor (3 A or more), 1 µF film capacitor (63 V), 1000 µF 25 V electrolytic capacitor, 4 Ω 5 W speaker.

Connections — do them in this order

1. Unplug the adapter first.
2. Inductor: one leg to column **SW**, the other leg to a new column called **OUT**. Its legs are thick and can loosen the breadboard holes, so twist or solder short thin wires onto them and plug those in.
3. **1 µF film capacitor from OUT to the blue rail (GND).**
4. **1000 µF capacitor:** + leg (long) to **OUT**. The stripe (–) leg to **speaker wire 1**.
5. **Speaker wire 2 → blue rail (GND).**
6. Replace the 10 Ω "safety" resistor with a plain wire (the speaker draws more current than the 10 Ω can handle).
7. Turn the phone volume to the **minimum**, then plug in the adapter, press play, and raise the phone volume slowly. Adjust the trimmer for a clean, loud sound.

TEST	Music plays from the speaker. The MOSFETs should be only warm, never too hot to touch. With the multimeter on DC volts, the voltage on the speaker terminal should be near **0 V** (not 6 V). If the speaker cone is pushed out and stays there, unplug at once: the DC-blocking capacitor is backwards or missing.

If the test fails: see the troubleshooting table on the next page.

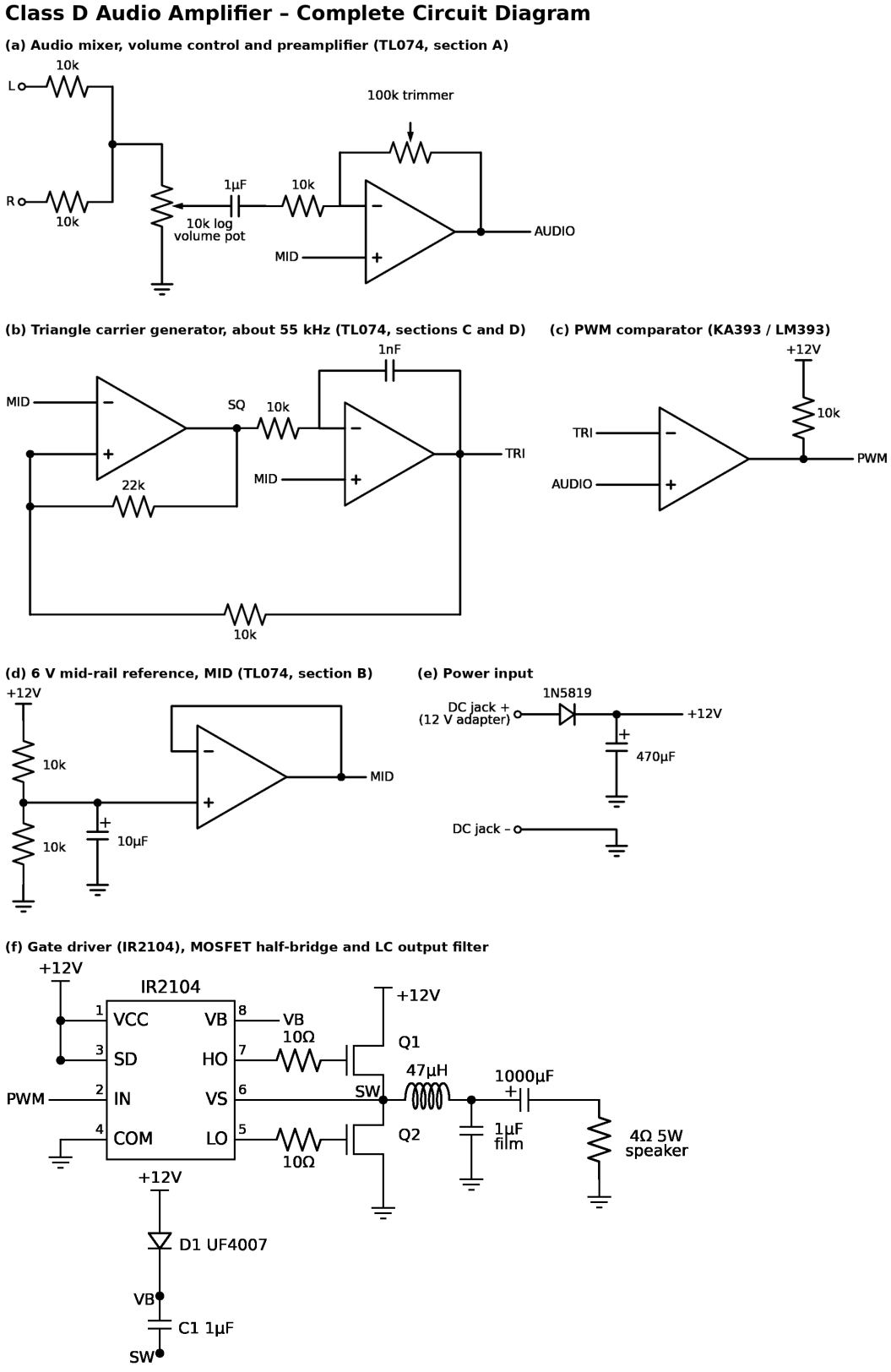
Voltage checklist (silent, no music)

Where to measure (against GND)	Expected reading
Top red rail (+12 V)	11 to 14 V
Bottom red rail (MID) and TL074 pin 7	about 6 V
TL074 pin 1 (audio out of the preamp)	about 6 V
TL074 pin 8 and pin 14 (oscillator)	about 6 V average (scope: triangle and square wave)
KA393 pin 1 (pulses)	about 6 V average (scope: 0 to 12 V pulses)
IR2104 pin 5	about 6 V average
Column SW	about 6 V average
Speaker terminal (after the 1000 µF)	near 0 V

Troubleshooting

Problem	Likely cause and fix
Nothing works, rail shows 0 V	Adapter polarity, diode backwards, or the 10 Ω resistor not in line. Re-do Stage 0.
No sound but all voltages look right	Phone volume at zero, wrong plug, the dongle not supported by your phone, or the pot is wired wrong. Check TL074 pin 1 with a scope or listen to the signal with headphones.
Sound is very quiet	Turn the trimmer to a higher resistance (more gain), raise the phone volume, check the pot wiring.
Sound is distorted or crackling	The audio is too loud for the amplifier. Lower the phone volume or the trimmer. Check that the triangle wave is clean at TL074 pin 8.
Humming or buzzing	Keep the audio input wires away from the MOSFET and inductor wires, use short ground wires, and add a 100 µF capacitor between +12 V and GND near the TL074.
MOSFET or resistor gets hot	Unplug immediately. Check that IR2104 pin 3 is joined to pin 1, the MOSFET legs are in the right columns, and there is no short between SW and a rail.
Loud pop when switching on	Normal for a few ms. Keep the phone volume low when you plug in the adapter.

Complete circuit (all stages together)



All GND symbols are common. KA393 pin 8 = +12 V, pin 4 = GND; TL074 pin 4 = +12 V, pin 11 = GND; 0.1 µF decoupling at every IC supply pin. Op-amp reference MID = 6 V. Carrier f ≈ 55 kHz; filter cut-off ≈ 23 kHz; ideal gain = 12 / (2 × 2) = 3.

Use this page to check your finished board. Pin numbers in blue are on the stage diagrams earlier in the guide.